**Case Study: School Threat Detection System**

## Business Case

The School Threat Detection System is an AI-powered security platform leveraging computer vision and object detection to monitor live camera feeds across school premises. The system identifies potential threats, such as weapons or suspicious objects, in real-time to enhance campus safety and enable rapid response. It aims to reduce incident response times, ensure compliance with privacy laws, and integrate with existing school security infrastructure.

### Scale

For scale, consider a school district with 50 schools, each equipped with 20–50 cameras, processing live feeds concurrently. The system must handle multiple schools, cameras, and alerts with high reliability and low latency.

### High-Level Goals

- **Outcome**: Accurate and timely detection of threats with minimal false positives, ensuring rapid alerts to security personnel.
- **Scalability**: Support multiple schools and cameras concurrently with modular, extensible architecture.
- **Quality**: High detection accuracy, reliable alert delivery, and robust error handling with retries for failed processes.
- **Security**: Ensure data privacy, compliance with regulations (e.g., FERPA, GDPR), and secure data transmission.
- **Multi-Tenant Support**: Logical separation of data for different schools or districts.
- **Observability & Reliability**: Real-time monitoring, logging, and rollback capabilities for failed detections or alerts.

## System & Architecture Design

Design a scalable, modular architecture for an AI-powered School Threat Detection System. The architecture must support multi-tenancy for multiple schools or districts and integrate with existing security systems.

Create a:

1. High-level system diagram.
2. List of key services and components.
3. Outline of data flow.

### System Requirements

- **Functional**:
  - Process live video feeds from multiple cameras in real-time.
  - Detect predefined threat objects (e.g., firearms, knives) using AI-based object detection.

- Send real-time alerts via email, SMS, or dashboard to designated personnel.
  - Provide a user interface for security personnel to monitor feeds and alerts.
  - Log all detections and alerts for auditing and compliance.

- **Non-Functional**:
  - Low latency: Detect threats and send alerts within seconds.
  - High availability: 99.9% uptime for continuous monitoring.
  - Scalability: Handle thousands of camera feeds across multiple schools.
  - Security: Encrypt video feeds and ensure compliance with privacy laws.
  - Reliability: Retry failed detections or alerts and maintain audit logs.

**Key Considerations**

- **Privacy**: Avoid facial recognition or individual identification to comply with ethical and legal standards.
- **Accuracy**: Balance sensitivity to detect threats with specificity to minimize false alarms.
- **Integration**: Seamlessly integrate with existing camera systems and security protocols.
- **Orchestration**: Manage distributed processing of video feeds across multiple cameras and schools.
- **Observability**: Provide dashboards and logs for monitoring system health and threat detection performance.